Author: That's right. Using recursion, let us try to determine one interesting sequence.

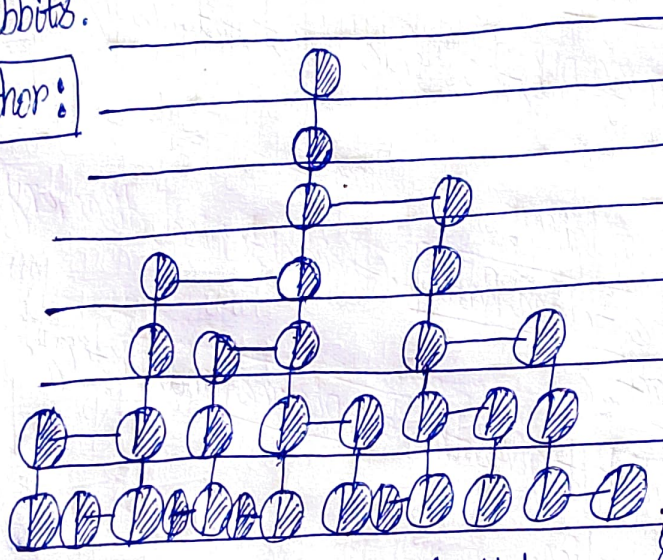
$$y_1 = 1;\ y_2 = 1;\ y_n = y_{n-2} + y_{n-1}$$

It's first terms are 1, 1, 2, 3, 5, 8, 13, 21, ... (11)

This sequence is known as the **Fibonacci seq/no.**

Reader: I understand, I have heard something about the problem of Fibonacci rabbits.

Author:



n	y_n
1	1
2	1
3	2
4	3
5	5
6	8
7	13

Symbol denotes one pair of rabbits

Yes, it was this problem, formulated by Fibonacci, the 13th century Italian mathematician, that gave the name to this sequence (11). The problem reads as follows. A man places a pair of newly born rabbits into a warren and wants to know how many rabbits he would have over a certain period of time. A pair of rabbits will start producing offspring two months after they were born and every following month one new pair of rabbits will appear. At the beginning (during the first month) the man will have in his warren only one pair of rabbits ($y_1 = 1$); during the second month he will have the same pair of rabbits ($y_2 = 1$); during the third month the offspring will appear, and therefore the number of the pair of rabbits in the warren will grow to two ($y_3 = 2$); during the fourth month there will be one more reproduction of the first pair ($y_4 = 3$); during the fifth month there will be offspring both from the first and second couples of rabbits ($y_5 = 5$), etc.

Reader: But in reality the rabbits do not multiply in accordance with such an idealized pattern. Furthermore, as time goes on, the first pairs of rabbits should obviously stop proliferating.